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A ADDITIONAL EXPERIMENTAL DETAILS AND RESULTS

A.1 MODEL ARCHITECTURE

Table 6 compares the architecture differences of the three models used in our experimental evaluation. All models supports the grouped query attention (GQA), in which multiple query heads share one KV head. Among them, the Yi-9B model has more transformer layers, while the Llama-3-8B model has more KV heads.

Table 6: Architecture overview of LLMs.

Model	Total Layer	Query Head	KV Head
Yi-6B	32	32	4
Yi-9B	48	32	4
Llama-3-8B	32	32	8

A.2 ADDITIONAL RESULTS ON NEEDLE-IN-A-HAYSTACK

Figure 7 shows the results of other methods on Needle-in-a-haystack benchmark. StreamingLLM can only find the correct answer when the needle’s position is within the static pattern. InfLLM maintains high performance with shorter context lengths. However, as the length increases, its performance shows a significant decline. Although SnapKV, Flat, and IVF perform well on this benchmark, we have analyzed their disadvantages in accuracy and latency in the previous evaluation.

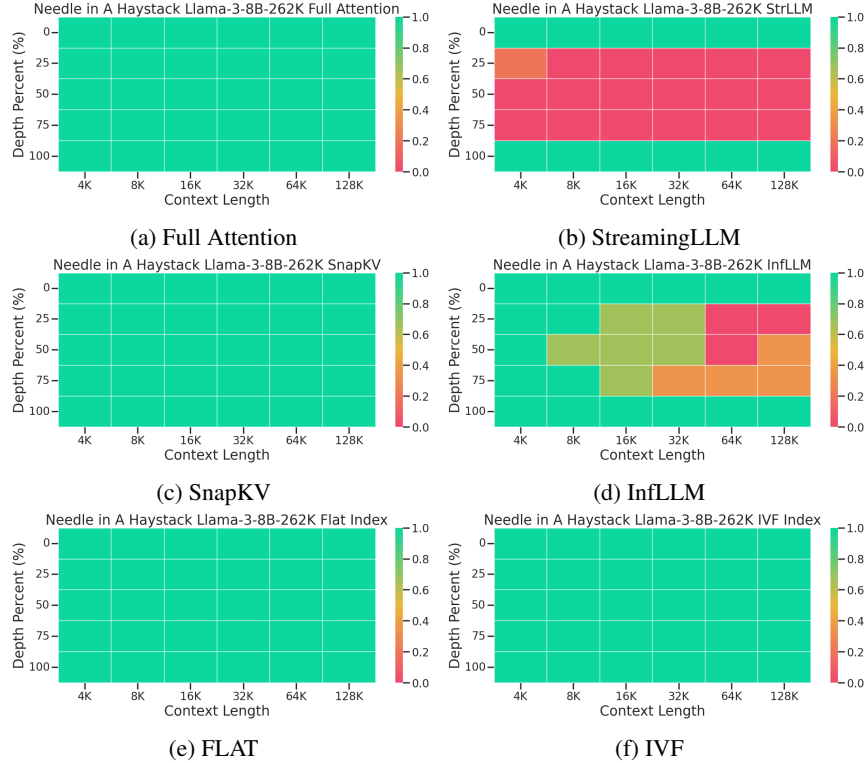


Figure 7: Performance of different algorithms and models on Needle-in-a-haystack. The size of the static pattern is consistently 640 (128 initial tokens + 512 tokens in the local window).